

Supplementary File 14. Evidence Gap Map

Purpose

This evidence gap map summarises where evidence is concentrated and where important gaps remain across the malaria and sickle cell disease (SCD) evidence base in sub-Saharan Africa. The map is based on the 68 included sources and the reconciled source-level coding from Supplementary Files 6 and 9.

The evidence gap map does not assess methodological quality. Its purpose is to show the volume, distribution, integration, and planning relevance of the evidence across the seven review domains.

Evidence Base Used for the Map

Item	Count
Included sources	68
Reconciled domain assignments	96
Evidence domains	7
Single-domain sources	39
Two-domain sources	21
Three-domain sources	4
High-integration sources	4

Reconciled Domain Counts

Domain code	Evidence domain	Sources coded
D1	Biological interaction between malaria and red blood cell polymorphisms	20

D2	Spatial co-risk and malaria-HbS/SCD geography	10
D3	Clinical convergence and under-recognition	10
D4	Service-mediated visibility and routine data	11
D5	Structural vulnerability and access to essential SCD services	20
D6	Programme response and screening-to-care systems	19
D7	Use of syndemic theory in malaria-SCD literature	6
Total domain assignments		96

1. Evidence Gap Map Coding Legend

Evidence level	Definition	Interpretation
Strong evidence concentration	15 or more sources in the domain or evidence area	Evidence is well represented and can support synthesis
Moderate evidence concentration	8 to 14 sources	Evidence is present but incomplete or uneven
Limited evidence concentration	3 to 7 sources	Evidence exists but remains thin or fragmented
Sparse evidence	1 to 2 sources	Evidence is highly limited
No direct evidence	0 sources	A clear evidence gap

2. Domain-Level Evidence Gap Map

Domain	Evidence volume	What is well represented	What remains weakly represented	Gap classification
D1 Biological interaction	20 sources	HbAS protection against severe malaria; HbS persistence; HbC; alpha-thalassaemia; G6PD deficiency; ABO; Dantu; epistasis; host genetic resistance	Translation of red cell polymorphism evidence into screening prioritisation, clinical algorithms, and service planning	Evidence strong, translation gap high
D2 Spatial co-risk	10 sources	Malaria mapping; HbS distribution; SCA birth burden; national sickle surveillance; routine malaria spatial methods	Integrated malaria-HbS/SCD co-risk maps linked to child population need, facility readiness, travel time, and referral systems	Evidence moderate, integration gap high
D3 Clinical convergence	10 sources	Severe anaemia; malaria-	Genotype-informed clinical	Evidence moderate, clinical

		positive illness; transfusion need; SCA under-recognition in severe anaemia admissions	attribution across HbSS, HbAS, and HbAA; malaria biomarkers in children with SCA; longitudinal outcomes	attribution gap high
D4 Service-mediated visibility	11 sources	Screening visibility; hospital genotyping; registries; mortality attribution; routine malaria data as methodological comparator	Routine SCD data systems; linkage between screening, clinics, admissions, transfusion, mortality, and follow-up	Evidence moderate, data-system gap high
D5 Structural vulnerability	20 sources	Mortality; prophylaxis; vaccination; hydroxyurea; transfusion relevance; comprehensive care needs; structural barriers	Facility readiness mapping; rural access; referral performance; laboratory monitoring; retention in decentralised systems	Evidence strong, implementation-readiness gap high

D6 Programme response	19 sources	Newborn screening; point-of-care testing; screening-to-care models; implementation networks; registry infrastructure	Complete cascade reporting from screening to survival; sustainability; financing; long-term retention	Evidence strong, cascade-completion gap high
D7 Syndemic theory	6 sources	Syndemic theory; biosocial framing; caution against co-occurrence-only claims	Empirical application of syndemic theory to malaria-SCD data; testing of interaction, structural amplification, and programme mediation	Evidence limited, empirical syndemic gap high

3. Matrix Evidence Gap Map

The matrix below shows how the 68 included sources distribute across the major evidence functions needed for a malaria-SCD syndemic evidence system.

Evidence domain	Biological mechanism	Spatial burden and co-risk	Clinical outcomes and	Diagnoses, visibility, and	Essential care and service	Programme implementation	Syndemic theory and
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Interpretation of the matrix

The map shows that evidence is strongest within domains rather than across domains. Biological evidence is strong, but it is rarely connected to service planning. Spatial evidence is moderate, but it is rarely connected to facility readiness or referral networks. Clinical evidence identifies severe anaemia as the main convergence syndrome, but genotype-informed attribution remains underdeveloped. Programme evidence is strong for screening feasibility and emerging implementation networks, but cascade completion and long-term retention remain less consistently documented.

The most important gaps are therefore not simple absence of malaria-SCD evidence. They are integration gaps across biological, spatial, clinical, data-system, service-readiness, and programme-response levels.

4. Cross-Domain Evidence Bridge Map

This map identifies where domains are already connected by included sources and where bridges remain weak.

Cross-domain bridge	Source count	Main supporting sources	Interpretation
D1 and D2: biological interaction plus spatial co-risk	2	S002, S008	HbS distribution is linked to malaria geography, but modern co-risk planning remains limited.
D1 and D5: inherited risk plus public health care needs	1	S062	Genomics and public health are linked, but biological evidence

			is rarely connected to service readiness.
D2 and D4: spatial burden plus service visibility	3	S025, S026, S027	Routine malaria data and screening surveillance show how geography and visibility interact.
D2 and D5: spatial burden plus care need	1	S010	SCA burden is linked to survival and service need, but facility readiness mapping remains limited.
D2 and D6: spatial surveillance plus programme response	1	S027	Sickle surveillance can support programme planning, but this bridge is underdeveloped.
D3 and D4: clinical convergence plus under-recognition	1	S035	This is a critical bridge showing hidden SCA among severe anaemia admissions.
D3 and D5: clinical vulnerability plus care need	4	S030, S033, S034, S040	Severe anaemia and SCD pathophysiology are linked to acute care and transfusion needs.

D4 and D5: hidden burden plus structural vulnerability	3	S028, S029, S048	Under-recognition and mortality attribution are linked to care access and follow-up.
D4 and D6: service visibility plus programme response	6	S027, S047, S048, S049, S051, S059	Screening, registries, and data systems make SCD visible for programme action.
D5 and D6: essential care plus programme response	8	S043, S044, S045, S046, S048, S052, S055, S060	Hydroxyurea, screening, care linkage, and programme systems are connected in several sources.
D7 with empirical domains	0 direct source-level overlaps	None	Syndemic theory is conceptually relevant but rarely integrated into empirical malaria-SCD studies.

5. Evidence Gap Priority Map

Priority level	Evidence gap	Why it is high priority	Linked domains
Critical	Lack of integrated malaria-HbS/SCD	Programmes cannot easily identify where	D1, D2, D4, D5, D6

	co-risk and service readiness mapping	high inherited risk, malaria exposure, weak services, and hidden burden converge.	
Critical	Under-recognition of SCA among children admitted with severe anaemia	Severe anaemia may be treated without identifying underlying SCA, delaying lifelong care.	D3, D4, D5
Critical	Incomplete screening-to-care cascade reporting	Screening cannot improve survival unless diagnosis is converted into continuous care.	D4, D5, D6
High	Weak routine SCD visibility and registry linkage	Routine data may underestimate burden where diagnosis and reporting are weak.	D2, D4, D6
High	Limited facility readiness evidence	Screening expansion may outpace capacity for confirmation, prophylaxis, hydroxyurea, transfusion, and referral.	D5, D6
High	Limited mortality attribution and cause-of-death linkage	SCD-attributable deaths may be recorded under	D3, D4, D5

		proximate causes such as infection, severe anaemia, or malaria-like illness.	
Moderate to high	Limited empirical syndemic modelling	The field has syndemic relevance but little empirical testing of interaction, structural amplification, and programme mediation.	D7 plus D1 to D6
Moderate	Limited evidence on long-term retention after diagnosis	Many screening studies report testing or enrolment, but fewer report sustained follow-up.	D5, D6
Moderate	Limited evidence on geographic access and referral completion	Travel time, catchment boundaries, and referral completion are rarely analysed with SCD burden.	D2, D5, D6

6. Evidence Gap Map by Research Question

Review question	Evidence available	Evidence gap	Priority output
How do malaria and red blood cell	Strong evidence for HbAS, HbC, alpha-thalassaemia, G6PD,	Limited translation of biological interaction	Biology-to-programme

polymorphisms interact biologically?	ABO, Dantu, and epistasis	evidence into service planning	translation framework
Where do malaria exposure and inherited sickle-related risk converge?	Moderate evidence from malaria maps, HbS maps, SCA birth estimates, and surveillance	Limited co-risk models that include facility readiness and access	Integrated malaria-HbS/SCD co-risk map
How do malaria and SCD converge clinically?	Moderate evidence on severe anaemia, malaria-positive illness, transfusion need, and SCA under-recognition	Limited genotype-informed attribution and outcome studies	Severe anaemia genotype and malaria biomarker study
How does SCD become visible or hidden in health systems?	Moderate evidence from screening, genotyping, registries, and mortality attribution	Weak routine SCD data linkage	Linked SCD visibility and routine data system
How do structural conditions shape outcomes?	Strong evidence that access to prophylaxis, hydroxyurea, transfusion, and comprehensive care matters	Limited facility readiness and geographic access evidence	Facility readiness and access assessment
What programme responses exist?	Strong evidence for screening, point-of-care testing, hydroxyurea, and implementation networks	Incomplete cascade and retention reporting	Standard screening-to-care cascade indicators

How is syndemic theory used?	Limited theory evidence and rare empirical use	Lack of empirical syndemic models	Bounded syndemic analytic framework
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